Radial rigidity for $\Delta u = f(|x|)\mathbf{1}_U - c\,\delta_0$

1 Statement

Let $U \subset \mathbb{R}^2 \simeq \mathbb{C}$ be a bounded open set with C^1 boundary, and let

$$f : (0, \infty) \to \mathbb{R}$$

be continuous with $f(r) > 0$ for every $r > 0$. Let $c \in \mathbb{R}$. Consider the distributional equation

$$\Delta u = f(|x|)\,\mathbf{1}_U - c\,\delta_0 \qquad \text{in } \mathcal{D}'(\mathbb{R}^2). \tag{1}$$

Theorem 1.1. *Assume $0 \notin \partial U$. Equation (1) admits a compactly supported distributional solution u with $u \in C^1(\mathbb{R}^2 \setminus \{0\})$ if and only if*

$$c = \int_U f(|y|)\,dA(y) \tag{2}$$

and U is rotationally invariant about the origin.

Remark 1.2. *A bounded rotationally invariant open subset of $\mathbb{R}^2$ is a disjoint union of (concentric) open annuli about the origin, together with at most one open disk centered at the origin. If one additionally assumes that U is connected, rotational invariance forces U to be a disk or a single annulus centered at the origin.*

2 The easy direction: radial U admits a compactly supported solution

Proposition 2.1. *Let $U = \bigsqcup_{j=1}^N A_j$ be a disjoint union of (concentric) open annuli or disks $A_j = \{r_j^- < |x| < r_j^+\}$ (with the convention $r_j^- \geq 0$; $r_j^- = 0$ gives a disk) centered at the origin, ordered so that $r_1^+ < r_2^- \leq r_2^+ < \cdots$. Assume f is continuous on $(0, \infty)$ and that (2) holds. Then (1) admits a compactly supported radial solution $u \in C^1(\mathbb{R}^2 \setminus \{0\}) \cap C^2(\mathbb{R}^2 \setminus (\partial U \cup \{0\}))$.*

Proof. We look for $u = u(r)$ depending only on $r = |x|$. Set

$$F(r) := \int_0^r s\,f(s)\,\mathbf{1}_U(s)\,ds,$$

so that F is continuous, nondecreasing, and $F(r) = \frac{1}{2\pi}c$ for all r larger than r_N^+ by (2).

Define, for $r > 0$,

$$g(r) := -\frac{c}{2\pi} + 2\pi \int_0^r s f(s)\mathbf{1}_U(s)\,ds \cdot \frac{1}{2\pi} = -\frac{c}{2\pi} + F(r),$$

and set

$$u'(r) := \frac{g(r)}{r}, \qquad u(r) := -\int_r^\infty \frac{g(s)}{s}\,ds.$$

The integral converges since $g(s) = 0$ for $s \geq r_N^+$ by (2). Moreover $u(r) = 0$ for $r \geq r_N^+$, so u is compactly supported.

For $r > 0$ we compute in the classical sense

$$\Delta u(r) = \frac{1}{r}\big(ru'(r)\big)' = \frac{g'(r)}{r} = f(r)\mathbf{1}_U(r) \qquad (r \in (0,\infty) \setminus \{r_j^\pm\}).$$

Across each $r = r_j^\pm$, g is continuous, hence u' is continuous; so $u \in C^1((0,\infty))$.

Near $r = 0$, $g(r) \to -c/(2\pi)$, so

$$u(r) = -\frac{c}{2\pi}\log r + O(1) \qquad (r \to 0^+).$$

Standard computation shows that the distributional Laplacian of u on $\mathbb{R}^2$ is

$$\Delta u = f(|x|)\mathbf{1}_U - c\,\delta_0,$$

since the radial profile $r \mapsto -\frac{1}{2\pi}\log r$ has distributional Laplacian δ_0. This proves the proposition. □

3 The hard direction: a compactly supported solution forces U to be radial

The proof follows the scheme suggested in `connected-annulus-rigidity.tex`, but we do *not* first show that the outer boundary is a circle. The crucial observation is that the vanishing of u in the unbounded exterior E_∞ already gives enough Cauchy data on Γ_{out} to kill the angular derivative of u across the whole connected component, irrespective of the shape of Γ_{out}.

Throughout this section assume:

(H1) $U \subset \mathbb{R}^2$ is bounded, open, with C^1 boundary, and $0 \notin \partial U$;

(H2) $f : (0,\infty) \to \mathbb{R}$ is continuous with $f(r) > 0$ for all $r > 0$;

(H3) $u \in C^1(\mathbb{R}^2 \setminus \{0\})$ is compactly supported and satisfies (1) distributionally.

3.1 Preliminary reductions

Denote by E_∞ the unbounded connected component of $\mathbb{R}^2 \setminus \overline{U}$, and by $\Gamma_{\text{out}} := \partial E_\infty$ the outer boundary of U. Since $0 \notin \partial U$, Γ_{out} is a compact subset of $\mathbb{R}^2 \setminus \{0\}$.

Lemma 3.1. *Under hypotheses (H1)–(H3), one has*

$$c = \int_U f(|y|)\,dA(y).$$

Proof. Test (1) with a cutoff $\chi \in C_c^\infty(\mathbb{R}^2)$ equal to 1 on a neighborhood of $\operatorname{supp}(u) \cup \{0\}$. Then $\int u\Delta\chi = 0$ since χ is constant on $\operatorname{supp}(u)$, and the right-hand side of (1) paired with χ gives $\int_U f - c$. □

Lemma 3.2. *Under (H1)–(H3), assume furthermore $0 \notin E_\infty$. Then*

$$u \equiv 0 \qquad \text{on } E_\infty.$$

Proof. On E_∞, the right-hand side of (1) vanishes: $\mathbf{1}_U = 0$ there, and δ_0 is not supported in E_∞ by hypothesis. Hence $\Delta u = 0$ in E_∞, so u is harmonic there (Weyl's lemma). Since u is compactly supported, u vanishes on the complement of a large ball, hence on a nonempty open subset of E_∞. As E_∞ is connected and u is real-analytic there, $u \equiv 0$ on E_∞. □

Remark 3.3. *If $0 \in E_\infty$, then on E_∞ we have $\Delta u = -c\,\delta_0$, so $u(x) = -\frac{c}{2\pi}\log|x| + h(x)$ with h harmonic on E_∞. But u is compactly supported, which forces $c = 0$ (by Lemma 3.1) and then $h \equiv 0$; the argument below then applies verbatim. We henceforth tacitly assume $0 \notin E_\infty$, which is the only nontrivial case.*

3.2 The angular derivative is harmonic

Let $L := x\partial_y - y\partial_x = \partial_\theta$ denote the generator of rotations, and set

$$w := Lu = \partial_\theta u \in C^0(\mathbb{R}^2 \setminus \{0\}).$$

Note that w need not be globally C^1 because L is only a first-order operator and u is a priori only C^1. Nevertheless, w is defined pointwise away from 0.

Lemma 3.4. *Under (H1)–(H3), w is harmonic in the open set $\Omega := \mathbb{R}^2 \setminus (\partial U \cup \{0\})$. In particular w is smooth in Ω.*

Proof. The operator L commutes with Δ and with multiplication by any radial function, and L annihilates both radial functions and δ_0. Therefore, in the interior of U (where $\mathbf{1}_U$ is locally constant $= 1$) and in the interior of $\mathbb{R}^2 \setminus \overline{U}$ (where $\mathbf{1}_U = 0$), away from the origin, we have

$$\Delta w = \Delta L u = L\Delta u = L\big(f(|x|)\mathbf{1}_U - c\,\delta_0\big) = 0.$$

Weyl's lemma upgrades the distributional identity to $w \in C^\infty(\Omega)$ harmonic. □

3.3 Cauchy data of w on the outer boundary

Lemma 3.5. *Under (H1)–(H3), and assuming $0 \notin E_\infty$:*

(a) $u \equiv 0$ and $\nabla u \equiv 0$ on Γ_{out};

(b) the extension $\tilde{w}$ of w by 0 to a two-sided C^1 neighborhood $\mathcal{N}$ of Γ_{out} (inside U on one side and inside E_∞ on the other) is continuous, with $\tilde{w} = 0$ on $\mathcal{N} \cap (E_\infty \cup \Gamma_{\text{out}})$.

Proof. (a) By Lemma 3.2, $u \equiv 0$ in E_∞, so the limiting values of u and ∇u from the exterior side of Γ_{out} vanish. Since $u \in C^1(\mathbb{R}^2 \setminus \{0\})$ and $\Gamma_{\text{out}} \subset \mathbb{R}^2 \setminus \{0\}$, the interior-side limits agree with the exterior-side ones, giving $u = 0$ and $\nabla u = 0$ on Γ_{out}.

(b) Because $\nabla u = 0$ on Γ_{out}, the angular derivative $w = x\partial_y u - y\partial_x u$ vanishes on Γ_{out}. On the other side, $w \equiv 0$ in E_∞ since $u \equiv 0$ there. Thus extending by 0 across Γ_{out} yields a continuous function $\tilde{w}$ on a neighborhood $\mathcal{N}$ of Γ_{out}. □

3.4 The key step: w vanishes near Γ_{out}

Lemma 3.6. *Under the standing hypotheses, the extension $\tilde{w}$ from Lemma 3.5(b) is harmonic on the neighborhood $\mathcal{N}$ of Γ_{out}. In particular*

$$\tilde{w} \equiv 0 \qquad \text{on } \mathcal{N}.$$

Proof. We already know $\tilde{w}$ is C^0 on $\mathcal{N}$ and smooth and harmonic on $\mathcal{N} \setminus \Gamma_{\text{out}}$ (using Lemma 3.4 and $\tilde{w} \equiv 0$ in $E_\infty \cap \mathcal{N}$).

For any test function $\varphi \in C_c^\infty(\mathcal{N})$, Green's formula applied on the two sides of Γ_{out} gives

$$\begin{aligned}
\int_{\mathcal{N}} \tilde{w}\,\Delta\varphi\,dA &= \int_{\mathcal{N}\cap U} w\,\Delta\varphi\,dA + \int_{\mathcal{N}\cap E_\infty} 0 \cdot \Delta\varphi\,dA \\
&= \int_{\mathcal{N}\cap U} \Delta w \cdot \varphi\,dA + \int_{\Gamma_{\text{out}}} \big(w\,\partial_\nu\varphi - \varphi\,\partial_\nu w\big)\,d\sigma \\
&= \int_{\Gamma_{\text{out}}} \big(w\,\partial_\nu\varphi - \varphi\,\partial_\nu w\big)\,d\sigma,
\end{aligned}$$

where ν is the outward unit normal to U and we used that w is harmonic in $\mathcal{N} \cap U$. The boundary integral vanishes: $w = 0$ on Γ_{out} by Lemma 3.5(b), and $\partial_\nu w = 0$ on Γ_{out} because $\nabla u = 0$ there (Lemma 3.5(a)) and w is a first-order linear combination of the components of ∇u whose coefficients depend only on x, y.

Therefore $\int \tilde{w}\,\Delta\varphi = 0$ for every $\varphi \in C_c^\infty(\mathcal{N})$, so $\Delta\tilde{w} = 0$ distributionally on $\mathcal{N}$. By Weyl's lemma, $\tilde{w}$ is a classical harmonic (hence real-analytic) function on $\mathcal{N}$. Since $\tilde{w}$ vanishes on the nonempty open set $\mathcal{N} \cap E_\infty$, the identity principle yields $\tilde{w} \equiv 0$ on $\mathcal{N}$. □

3.5 From local vanishing to global rigidity

Lemma 3.7. *Under (H1)–(H3), $w \equiv 0$ in every connected component V of U whose closure meets Γ_{out}. In particular, if U is connected, $w \equiv 0$ in $U^\circ \setminus \{0\}$.*

Proof. Let V be such a component. Lemma 3.6 gives $w \equiv 0$ on the open set $V \cap \mathcal{N} \neq \emptyset$. The set $V \setminus \{0\}$ is connected (a connected planar open set remains connected after removing a point), w is harmonic, hence real-analytic, on $V \setminus \{0\}$, and vanishes on a nonempty open subset. By the identity principle, $w \equiv 0$ on $V \setminus \{0\}$, hence on V by continuity away from 0. □

Lemma 3.8. *Under (H1)–(H3), $w \equiv 0$ on all of $U \setminus \{0\}$.*

Proof. Call a connected component V of U *reached* if $w \equiv 0$ on $V \setminus \{0\}$. Lemma 3.7 shows that every component of U meeting the unbounded exterior E_∞ through a common boundary arc is reached. We now propagate this to the other components.

Let V be any component of U, and let W be the connected component of $\mathbb{R}^2 \setminus \overline{V}$ that contains V's outer boundary, i.e., W is the unbounded component of the complement of V. If $W \supset E_\infty$, then V's outer boundary is part of Γ_{out} and V is reached by Lemma 3.7.

Otherwise W is a bounded component of $\mathbb{R}^2 \setminus \overline{U}$ nested inside another component V' of U. We proceed by induction on the nesting depth (which is finite since U has finitely many connected components: any bounded open set with C^1 boundary has finitely many components because ∂U is a finite disjoint union of compact C^1 curves). The inductive hypothesis is that the parent component V' is reached, i.e., $w \equiv 0$ on $V' \setminus \{0\}$. Hence $\partial_\theta u = 0$ on $V' \setminus \{0\}$ up to $\partial V'$, and in particular $u|_{V'}$ is radial up to the boundary it shares with $\overline{W}$. We cannot yet conclude $u \equiv 0$ on W, but we will invoke a different argument: we show that w also vanishes on W, by applying Lemma 3.6 with E_∞ replaced by W, using that $w = 0$ on the "outside" of W (which is V') via the inductive hypothesis, and that $w = \partial_\theta u$ has vanishing Cauchy data on ∂W because ∇u is continuous across ∂W ($u \in C^1(\mathbb{R}^2 \setminus \{0\})$) and $\partial_\theta u = 0$ on V'. The same reflection/identity-principle argument then gives $w \equiv 0$ on a neighborhood of ∂W on the W-side. Proceeding component by component down the nesting tree, $w \equiv 0$ on every component V of U. □

Remark 3.9. *If one assumes U connected (as in* `connected-annulus-rigidity.tex`*), the "propagation" step in Lemma 3.8 is unnecessary and the proof collapses to Lemma 3.7.*

3.6 Conclusion

Proof of Theorem 1.1, $\Rightarrow$ direction. Combining Lemmas 3.1 and 3.8, we have $c = \int_U f\,dA$ and $\partial_\theta u \equiv 0$ on $U \setminus \{0\}$, so u is radial on $U \setminus \{0\}$. Also $u \equiv 0$ on E_∞, which is radial (as complement of a relatively closed set, once we know U is radial; here we simply note $u|_{E_\infty} = 0$ is radial trivially). In any bounded component of $\mathbb{R}^2 \setminus \overline{U}$ disjoint from the origin, u is harmonic; as a harmonic function whose angular derivative vanishes on the adjacent component of U (hence on a full open arc of the boundary of the bounded component) and which is C^1 across that boundary, u is also radial there by the same Cauchy uniqueness argument as above.

Thus u is radial on $\mathbb{R}^2 \setminus \{0\}$.

Returning to the distributional equation (1): both Δu and $c\,\delta_0$ are rotationally invariant distributions (rotation invariance commutes with Δ), so

$$f(|x|)\mathbf{1}_U = \Delta u + c\,\delta_0$$

is rotationally invariant as a distribution. Because $f(|x|)$ is a strictly positive continuous function on $(0, \infty)$ and is itself rotationally invariant, the product $f(|x|)\mathbf{1}_U$ being rotationally invariant forces $\mathbf{1}_U$ to be rotationally invariant as a locally integrable function. Concretely, for any rotation R_θ about the origin, the identity $f(|x|)\mathbf{1}_U(x) = f(|x|)\mathbf{1}_U(R_\theta x)$ holds almost everywhere

in $\mathbb{R}^2 \setminus \{0\}$, and since $f(|x|) > 0$ on that set, $\mathbf{1}_U(x) = \mathbf{1}_U(R_\theta x)$ almost everywhere. Equivalently, the symmetric difference $U \triangle R_\theta U$ has Lebesgue measure zero; since both sets are open, this symmetric difference is open with empty interior, hence empty. Therefore $U = R_\theta U$ for every rotation, i.e., U is rotationally invariant about the origin. □

4 The connected case

Corollary 4.1. *Under the hypotheses of Theorem 1.1, if U is connected then U is either a disk or an annulus centered at the origin.*

Proof. A bounded connected open subset of $\mathbb{R}^2$ that is rotationally invariant about the origin is determined by its intersection with a ray from the origin; this intersection must be an open connected subset of $(0, \infty)$ (possibly with 0 included, if $0 \in U$), i.e., an open interval. Rotating gives either an open disk $\{|x| < R\}$ (if $0 \in U$) or an open annulus $\{r_1 < |x| < r_2\}$ (if $0 \notin U$). Theorem 1.1 gives the rotational invariance; the shape classification follows. □

5 Why this works where `main.tex`'s simply connected argument did not

The Section 4 argument of `main.tex` constructs a holomorphic function W on U whose boundary values are real, then invokes the maximum principle for $\Im W$ inside a *simply connected* U to conclude $\Im W \equiv 0$. In multiply connected U, the values of $\Im W$ on inner boundary components are not controlled and the maximum principle argument fails — this is the obstruction identified in the remark at the end of `connected-annulus-rigidity.tex`.

Our proof sidesteps this by working with the real, first-order, rotation-generator $L = \partial_\theta$ rather than with the complex, holomorphic object W:

- Lu is harmonic in U for purely algebraic reasons (rotation invariance of the right-hand side away from ∂U);
- the Cauchy data of Lu on *any* outer boundary component (not just a circle) is immediately zero because u vanishes together with its gradient on Γ_{out};
- a harmonic function with zero Cauchy data on a portion of a C^1 boundary is zero in a neighborhood of that portion (Lemma 3.6), by extension-by-zero and the identity principle;
- connectedness of $U \setminus \{0\}$ (which only requires U connected and finite, not simply connected) then propagates this vanishing throughout U.

No assumption that Γ_{out} is a circle is needed; the conclusion "Γ_{out} is a circle" is a consequence, not a hypothesis.

Remark 5.1. *The argument uses* only *rotational symmetry of the right-hand side, C^1 regularity of ∂U, and vanishing of u in the unbounded exterior (which comes for free from compact support). It does not use any moment identities, holomorphic machinery, Bargmann/Hermite structure, or free-boundary theory. In particular, it gives the full rigidity theorem for* every *continuous positive radial weight f, not only for Hermite weights.*